

Initialisation

Split months by volatility $\rightarrow$ starting guesses for $\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k, \mathbf{P}$

Repeat for $M = 2,000$ iterations (first $B = 500$ discarded as burn-in)

Step 1: Assign each month to a regime (FFBS)

$$P(s_t \mid \mathbf{z}_{1:T}) \propto f(\mathbf{z}_t \mid s_t) \cdot P(s_t \mid s_{t-1})$$

posterior $\propto$ likelihood $\times$ transition prior

Holds $\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k, \mathbf{P}$ fixed

Step 2: Learn what each regime looks like (NIW)

$$p(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k \mid \mathbf{z}_k) \propto \prod_{t: s_t = k} \mathcal{N}(\mathbf{z}_t; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \cdot \text{NIW}(\mathbf{m}_0, K_0, \nu_0, \boldsymbol{\Psi}_0)$$

posterior $\propto$ likelihood $\times$ conjugate prior

Holds $\{s_t\}$ and $\mathbf{P}$ fixed

Step 3: Learn how often regimes switch (Dirichlet)

$$p(\mathbf{P}_{i, \cdot} \mid \{s_t\}) \propto \prod_t P_{i, s_t}^{n_{ij}} \cdot \text{Dir}(\boldsymbol{\alpha}_i)$$

posterior $\propto$ likelihood $\times$ conjugate prior

Holds $\{s_t\}, \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k$ fixed

next
iteration

Output

Average the post-burn-in draws: $\bar{\boldsymbol{\mu}}_k, \bar{\boldsymbol{\Sigma}}_k, \bar{\mathbf{P}}$

Apply forward filter $\rightarrow \pi_t^{\text{filter}} = P(s_t = 1 \mid \mathbf{z}_{1:t})$